

Agentic AI and Wireless Applications

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Sep. 2026



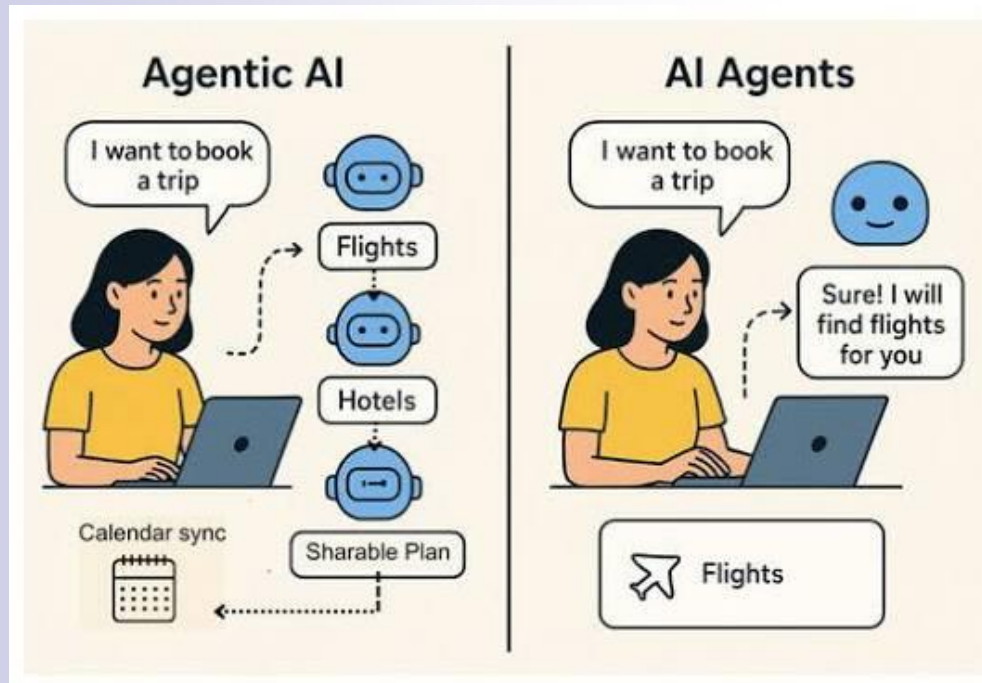
Outline

- ❑ Background: Agentic AI & Applications
- ❑ Preliminaries
- ❑ Constrained Multi-Agent Reinforcement Learning with Radio Map Estimation
- ❑ Theoretical Analysis
- ❑ Results
- ❑ Future Direction

Agentic AI

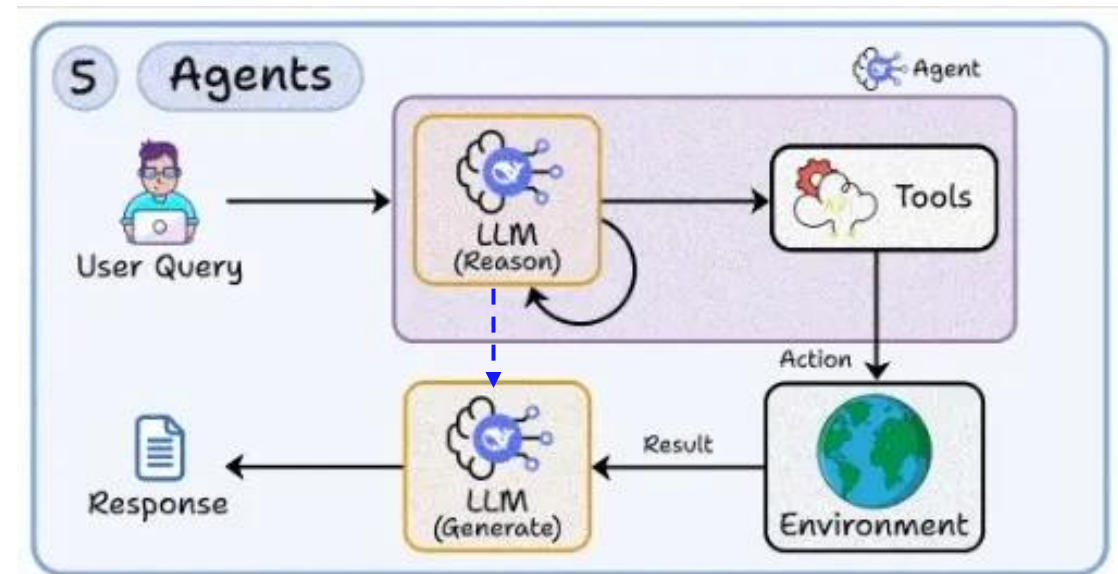
❑ Agentic AI vs. AI agent

An agentic AI tool runs a perception-reasoning-planning loop for trustworthy, general and self-adaptive decision making for services.



Skills vs. Prompt

- ✓ **Autonomous:** Acts without a human in the loop.
- ✓ **Goal-directed:** Optimizes a serving objective.



Agentic Structure

- ✓ **Closed-loop adaptivity:** self-reflection.
- ✓ **Long-horizon delegation:** many steps accomplish.

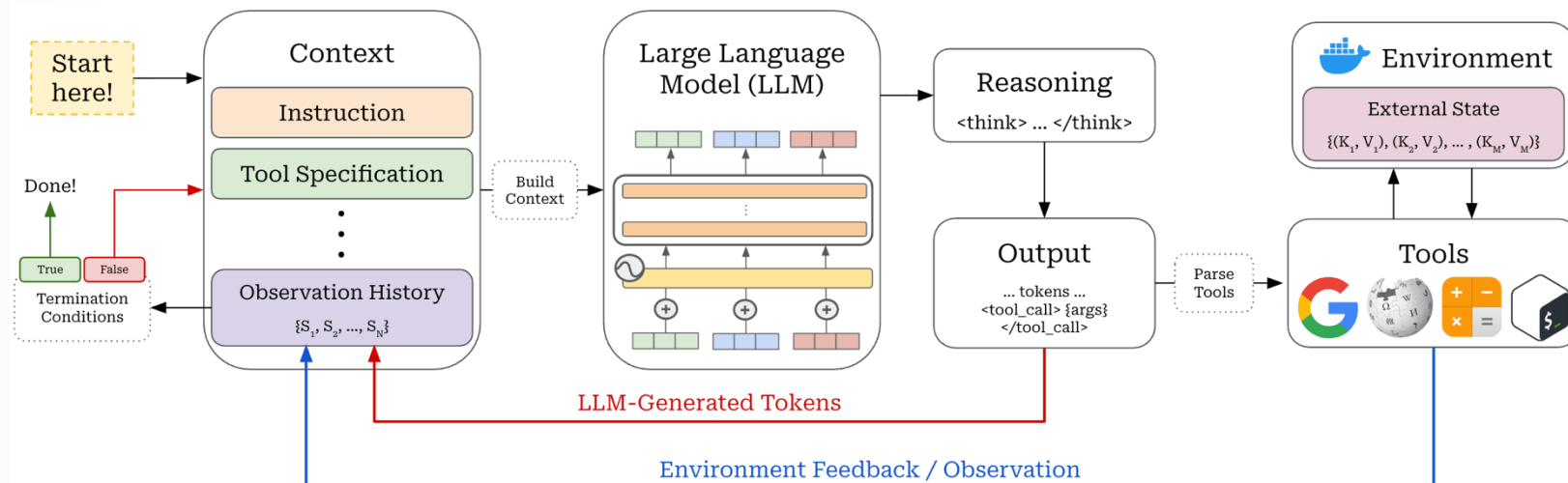
Agentic RL

❑ Post-training LLM chat as RL

- Task decoupling.
- Tool use scheduling.
- HFRL for policy updates.
- Response after multi-turn rollout.



- ✓ **Deep thinking & searching**
- ✓ **Tool-integrated reasoning**
- ✓ **Memory based reflection**



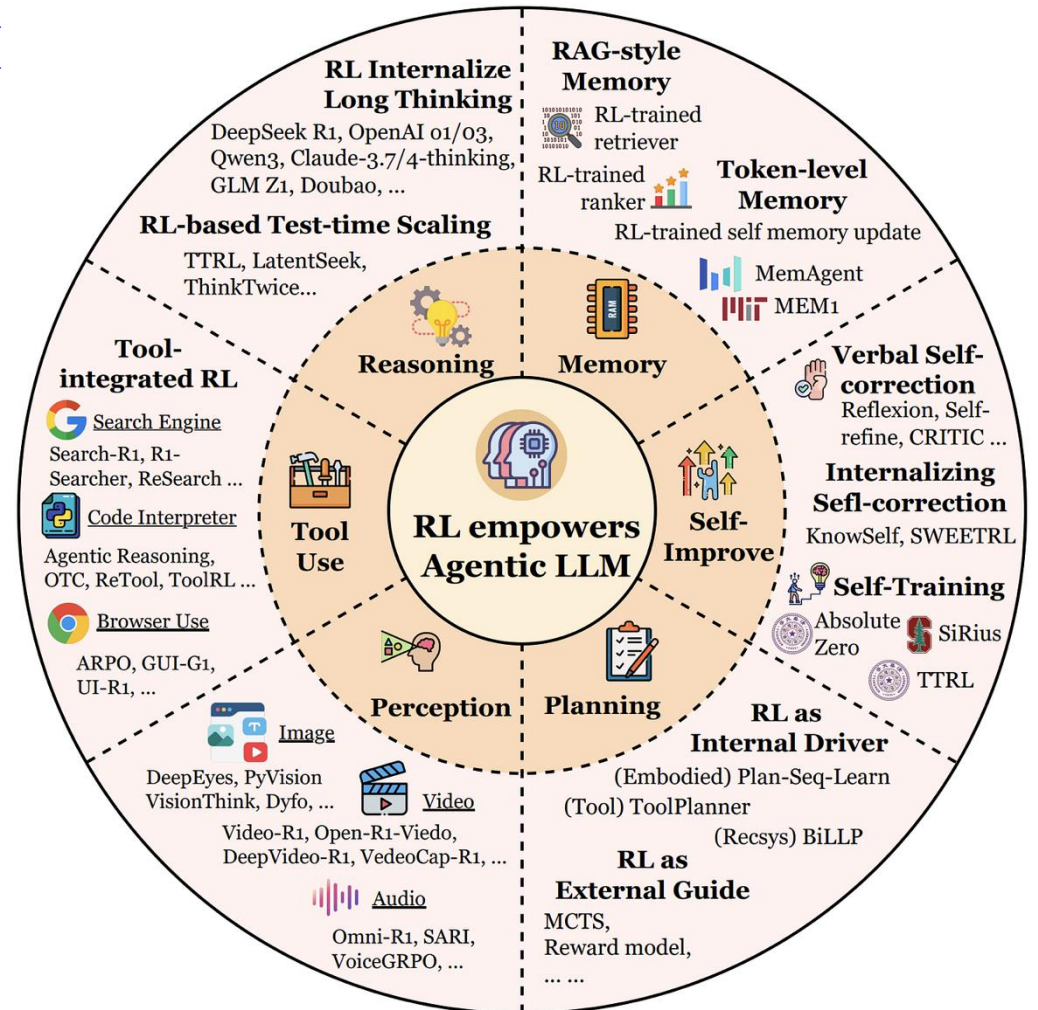
Agentic RL

❑ Agentic RL Forging the Next Generation of AI

Agentic RL: memory-based evaluation

- Task-oriented perception
- Autonomous tool using
- Self-adaption and correction

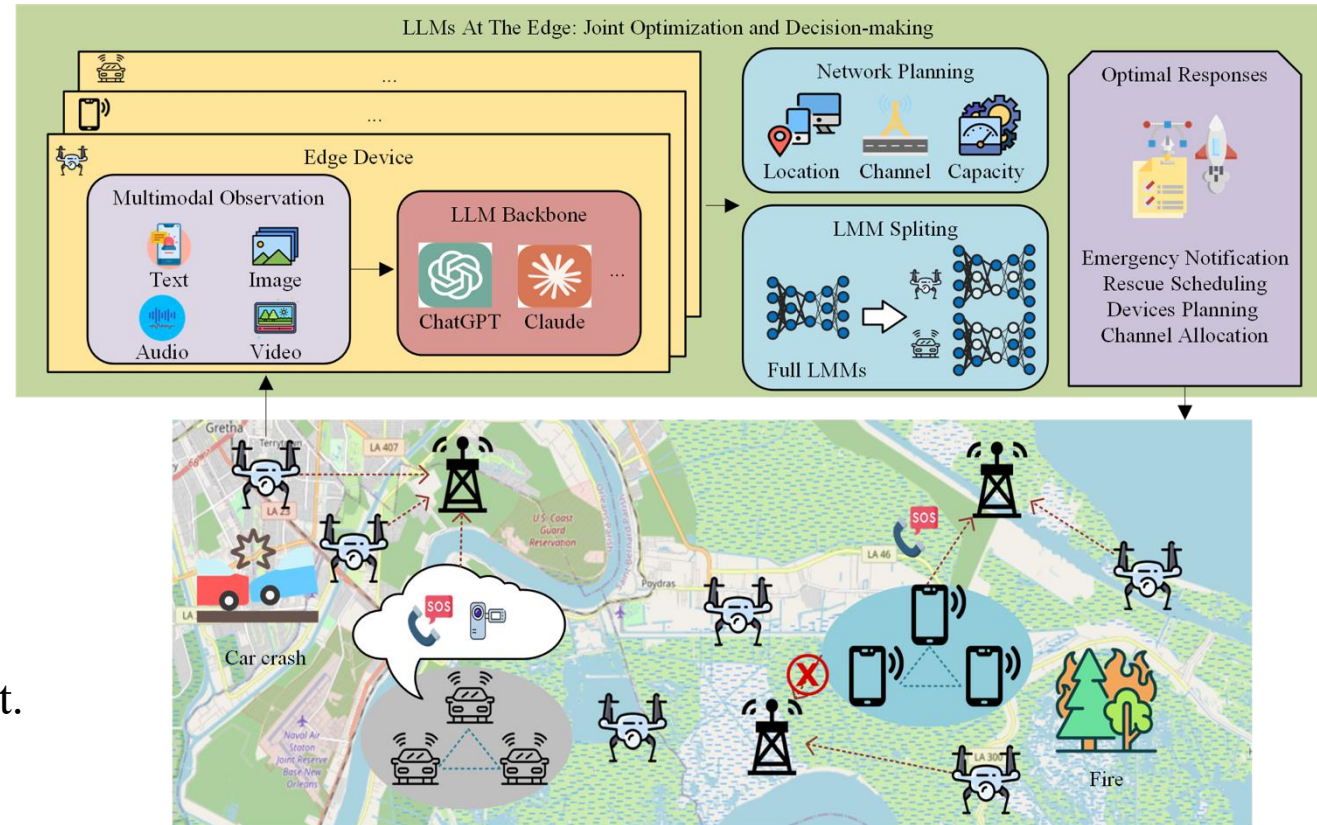
- ✗ Sample efficiency
- ✗ Dynamic environment
- ✗ Long term rollout & Sparse Reward
- ✗ Heterogeneous & partial observations



Agentic AI in Wireless Systems

❑ Agentic AI is suitable for wireless

- Multi-objective downstream tasks
- Complex constraints
- Dynamic wireless environment



- ✓ **Generalization:** outperforms in various scenarios.
- ✓ **Trustworthy decision-making:** multiple turns trajectory reasoning and planning before deployment.
- ✓ **Scalability:** multi-tasks with priority design.
- ✓ **Self-adaption:** tool use, adaptive sensing without human intervention.

Motivation & Challenge

□ A general UAVs control system for reliable spectrum awareness and efficient services.

➤ Example: Edge devices for mail delivery.

❖ No global wireless observation, no well-designed RME model

➤ Challenges:

❖ Distribution shift under unseen regions

❖ Efficiency and effectiveness

❖ Safe collaboration

❖ Perception & control scheduling

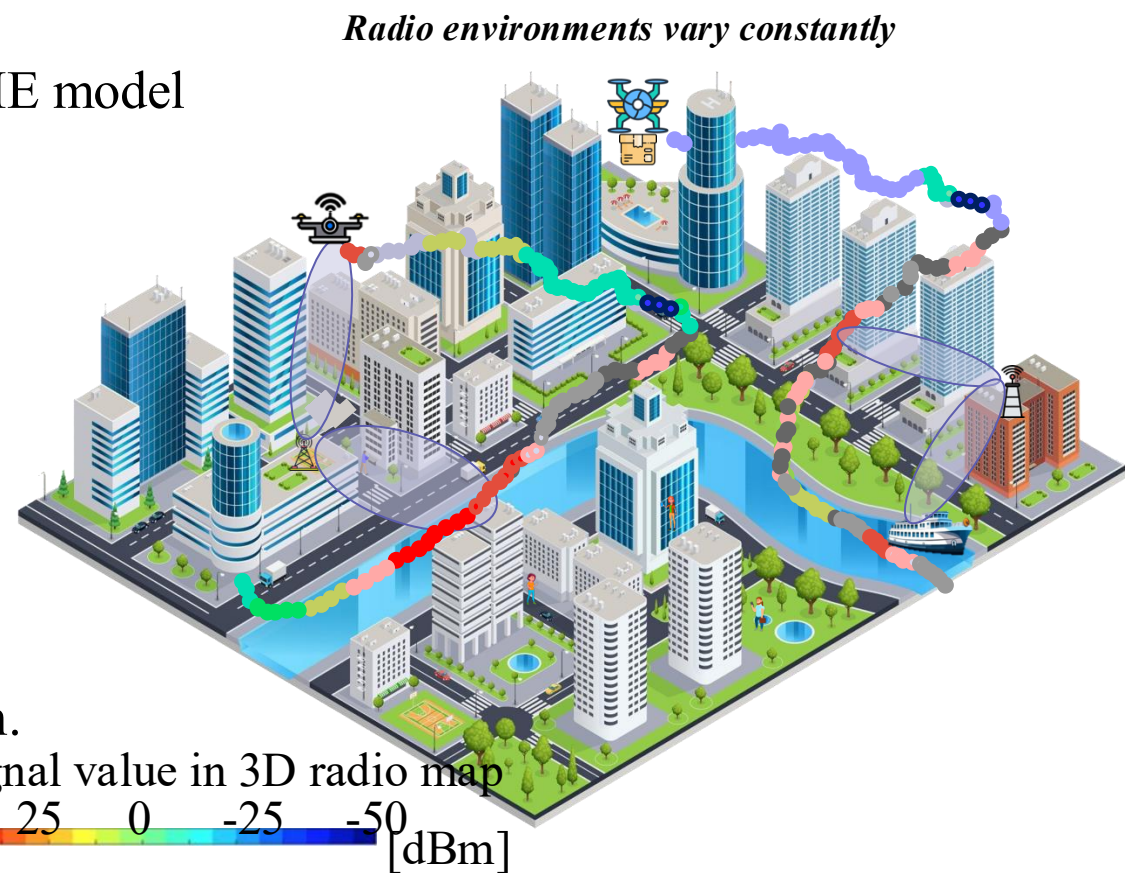
➤ Contributions:

C-MARL for navigation & RME fine-tuning

❖ **Constraint-MARL:** Env. Setting & constraint design.

❖ **Credit assignment:** Efficient policy updates.

❖ **Gated test-time training:** Reliable RME evaluation and effective fine-tuning trigger



Preliminaries

□ Wireless

Devices keep a reliable awareness and stable connection while serving.

➤ Awareness: the acc\uncertainty of RME prediction.

❖ Pre-trained VAE radio map estimation for

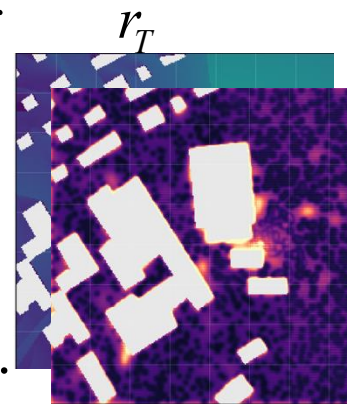
$$\Theta : \{o^s\} \rightarrow r \sim \mathcal{N}(r_{acc}, r_{var})$$

➤ Outage: the bounded outage probability^[1].

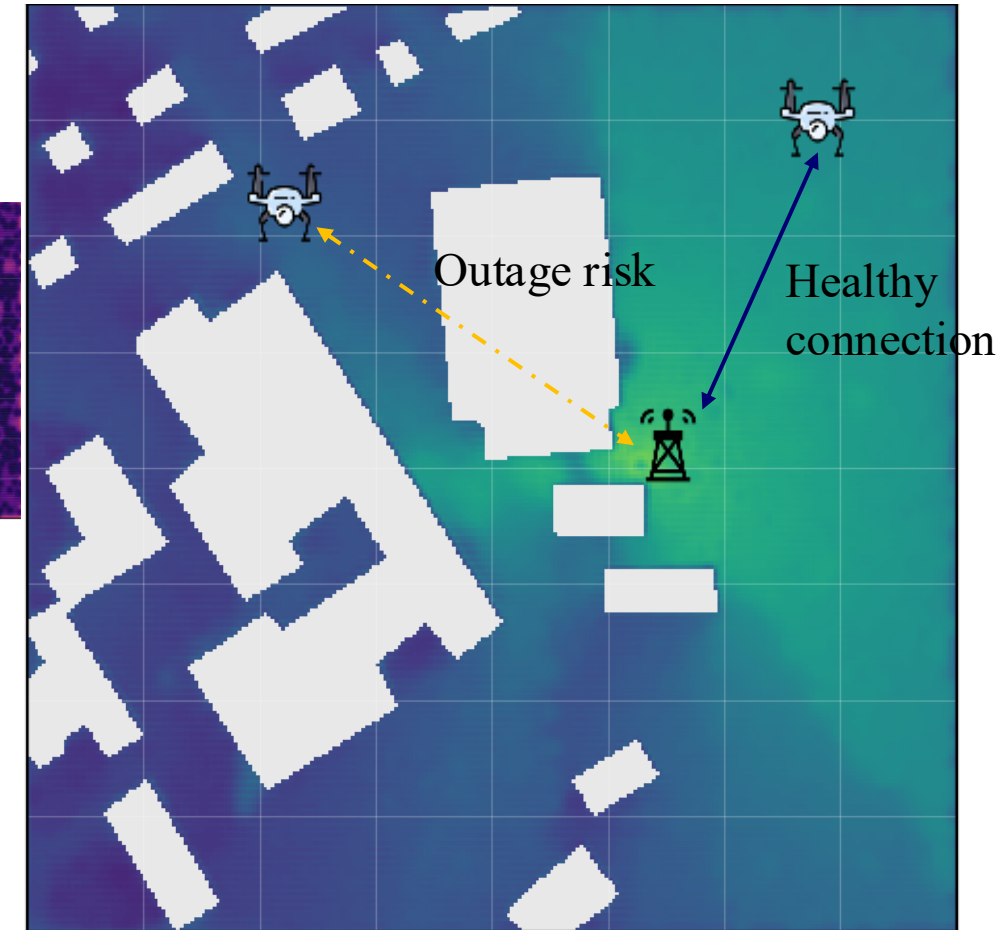
❖ Gird region based outage

$$\mathbf{G}^{N \times N} = \{\mathbf{g}_i\}_0^{N^2-1}$$
$$p_i^{\text{out}} \geq \Gamma_o, \quad p_i^{\text{out}} = \frac{1}{|\mathbf{g}_i|} \sum_{\mathbf{g}_i} \Pr(r < \phi),$$

$$u_i = \frac{1}{|\mathbf{g}_i|} \sum_{\mathbf{g}_i} r_{var}.$$



Radio map



[1] Andrews, Jeffrey G., François Baccelli, and Radha Krishna Ganti. "A tractable approach to coverage and rate in cellular networks." *IEEE Transactions on communications* 59.11 (2011): 3122-3134.

Preliminaries

□ Reinforcement Learning

- Decentralized partially observable Markov decision process (Dec-POMDP)

$$\pi_{\theta} = \arg \min J(\langle U, S, \{A\}, R, \{O\}, P, \gamma \rangle)$$

- ❖ Value function $v_{\pi}^{\theta, \tau}(s_t) = \sum_{a_t} \pi(a_t | s_t) \left(R_{a_t, s_t} + \gamma \sum_{s_{t+1}} P(s_{t+1} | a_t, s_t) v_{\pi}(s_{t+1}) \right)$

$$q_{\pi}(a_t, s_t) = R_{a_t, s_t} + \gamma \sum_{s_{t+1}} P(s_{t+1} | a_t, s_t) \sum_{a_{t+1}} \pi(a_{t+1} | s_{t+1}) q_{\pi}(a_{t+1}, s_{t+1})$$

- ❖ General advantage estimation

$$\hat{A}_t(\gamma, \psi) = \sum_i (\gamma \psi)^i (r_t + \gamma V(s_{t+1}) - V(s_t)).$$

- Q-Learning & DDPG: build and update Q-table

$$Q^*(a_t, s_t) = \mathbb{E}_{s_{t+1} \sim P} \left[r(a_t, s_t) + \gamma \max_{a_{t+1}} Q^*(a_{t+1}, s_{t+1}) \right]$$

- PPO: policy gradient

$$J_{PPO}(\theta) = \frac{1}{N} \sum_{i=1}^N \min \left(\frac{\pi_{\theta}(a_t^{(i)} | s_t)}{\pi_{\theta_{old}}(a_t^{(i)} | s_t)} A(s_t, a_t^{(i)}), \text{clip} \left(\frac{\pi_{\theta}(a_t^{(i)} | s_t)}{\pi_{\theta_{old}}(a_t^{(i)} | s_t)}, 1 - \delta, 1 + \delta \right) A(s_t, a_t^{(i)}) \right)$$

Methodology

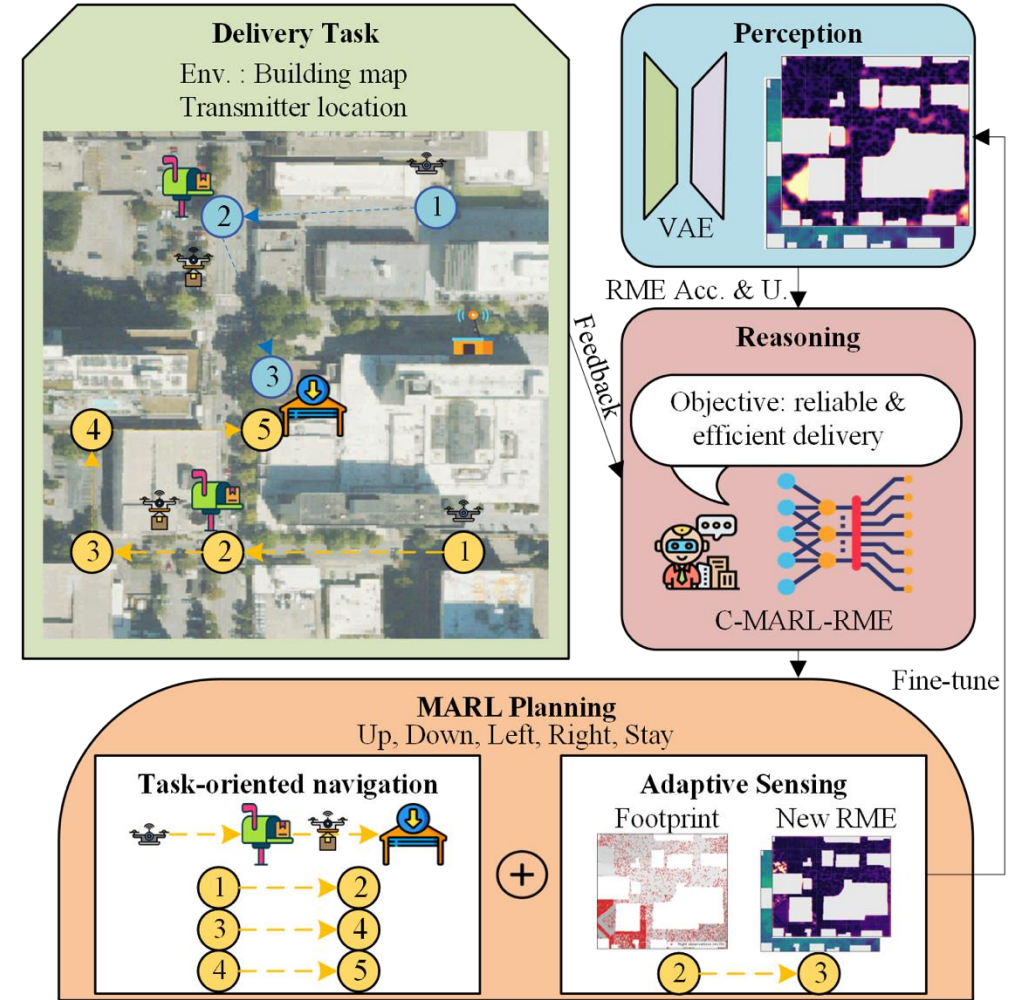
□ System Overview

- Target: deliver with bounded wireless awareness
- Algorithm deployment
 - ❖ Edge devices: C-MARL, VAE, sensing
 - ❖ Server: VAE fine-tuning & evaluation
- Serving in unseen region: delivery and sensing
 - ❖ Perception: enrich observations for fine-tuning
 - ❖ Reasoning: update policy for serving
 - ❖ Planning: adaptively sensing or navigation

$$\min(T + \mu T_{\text{out}} + \omega e_{\text{out}}) \rightarrow \max_{\theta} E_{\tau \sim \pi_{\theta}} \left[\sum_{t=0}^{T-1} \gamma^t r_t \right]$$

$$T_{\text{out}} = \sum_{t=1}^T \sum_{i=1}^U \Pr(p_{x_i,t}^{\text{out}} \geq \Gamma_o), \quad e_{\text{out}} = \sum_{i=0}^{N^2-1} \left| p_{i,T}^{\text{out}} - p_{i_T,T}^{\text{out}} \right|$$

$$\text{s.t.} \quad E_{\tau \sim \pi_{\theta}} [C(\tau)] \leq \Gamma_c.$$



Constraint Design

□ C-MARL

➤ Wireless constraint: uncertainty is declined while mission completed.

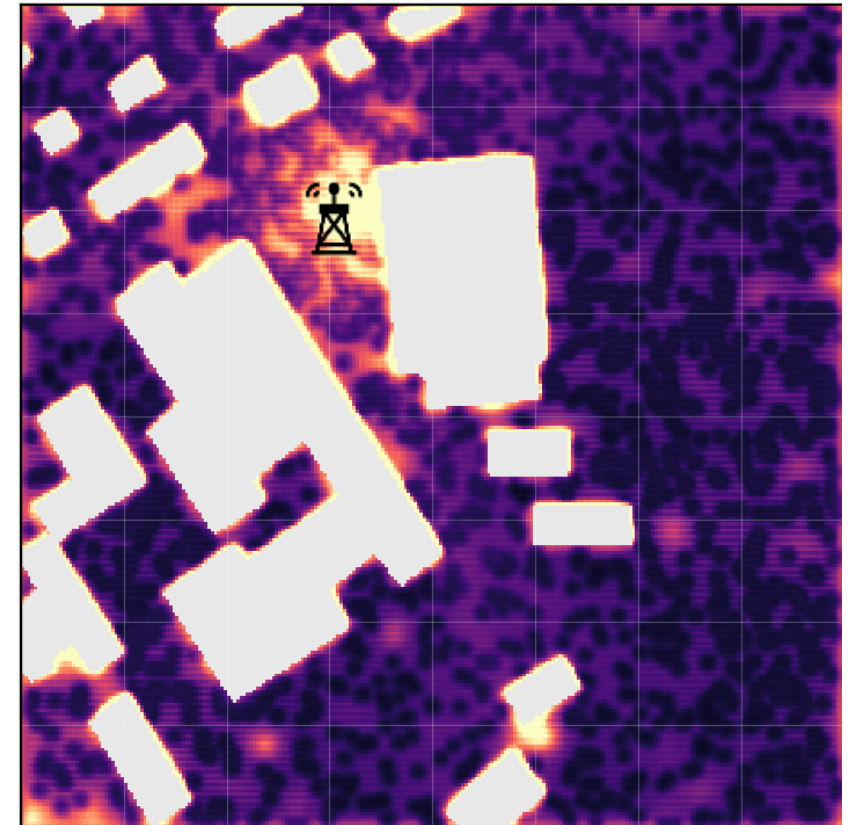
❖ Inaccurate RME under distribution shift

$$u_{i,T} \leq \Gamma_a, \quad \forall i \in R_i^h, R_i^h = \{g_i : u_{i,0} > \Gamma_a\}$$

❖ Challenges around building and Tx

$$s.t. \quad \sum_{i \in R_i^h} \max(0, u_{i,T} - \Gamma_a) \leq \Gamma_c$$

$$R_i^h = \{g_i : u_{i,0} > \Gamma_a\}, \quad \Gamma_a = \max(\Gamma_u, k u_{i,0})$$



Uncertainty map

Environment

□ C-MARL: Dec-POMDP

- State: RME model, wireless knowledge, tasks based knowledge

$$s_t = \{\Theta_t, s_t^w, s_t^d\}, s_t^w \leftarrow \{r_T\}, s_t^d = \left\{ \{\{x_{i,t}\}^U, \{d_i\}^m, d_p\}_0^{N^2-1}; \{c_{i,t}\}^U; \{m_t, p_t, a_t\}^m \right\}$$

- Observation: local sensing, global delivery observation (assumption)

$$o = \{r_t, p^{\text{out}}, u, R^h, o^s, o^d\}, o^s = \{o_f^s, o_s^s\}$$

- Action: navigation and sensing behavior includes observation, as sensing observe more

$$a_t \in \underbrace{\{\text{UP}, \text{DOWN}, \text{LEFT}, \text{RIGHT}\}}_{\text{navigation}}, \underbrace{\{\text{STAY}\}}_{\text{sensing}}$$

$$|o_s^s| = 4|o_f^s|$$

Solution

□ C-MARL

➤ Lagrangian Relaxation

$$\max_{\theta, \lambda} \mathbb{E}_{\tau \sim \pi_{\theta}} \left[\sum_{t=0}^{T-1} \gamma^t r_t \right] - \lambda \left(\mathbb{E}_{\tau \sim \pi_{\theta}} [C(\tau)] - \Gamma_c \right)$$

$$\lambda \leftarrow \text{clip}(\lambda + \eta_{\lambda} (\bar{c} - \Gamma_c), 0, \lambda_{\text{cap}})$$

➤ Reward and cost channel

$$r_t = -1 - \mu \sum_i p_{x_{i,t}}^{\text{out}} + \sum_t [\gamma \Phi_d(s_{t+1}) - \Phi_d(s_t)]$$

$$c_t = \omega e_{\text{out}} + \sum_{i \in R^h} \max(0, u_{i,T}^{\text{cal}} - \Gamma_a)$$

➤ Implementation tricks

- ❖ Memory based uncertainty estimation $u_{i,T}^{\text{cal}} = u_{i,T} \cdot \text{clip}(\max(1, \bar{z}_i^2), 1, \Gamma_m)$
- ❖ Adaptive lambda schedule: loose constraint at the beginning

RME Fine-Tuning

Pre-trained VAE degrades with distribution shift.

- Trigger: local evaluation by o_s^s

$$T_i: u_i^{\text{cal}} > \Gamma_a \quad \text{or} \quad |\bar{q}_i - p_{i,T}| > \kappa_z \sqrt{u_i^{\text{cal}}}$$

- Adaptive sensing: one stay action

- Fine-tuning: $\Theta_{\text{new}} = \arg \min_{\Theta} L_{\text{recon}}(\Theta; P_{\text{mem}}) + \beta \|\Theta - \Theta_{\text{ref}}\|^2$

- Accept gate: does fine-tuned model benefit under heterogeneous and sparse observation?

- ❖ The disjoint inference and evaluation set $P_{\text{mem}} \cup V = \{o_f^s, o_s^s\}$
- ❖ Synchronous evaluation, asynchronous fine-tuning

Algorithm 1: Accept Gate under Distribution Shift

Require: Θ_{new} ; pre-update Θ_t ; V ; margin $\varepsilon_{\text{gate}} \geq 0$; minimum size V_{min} ; decay ρ ; residuals $\{\bar{z}_i^2\}$

Ensure: model updates Θ_{t+1} ;

```

1: if  $|V| < V_{\text{min}}$  then
2:   return  $\Theta_{t+1} \leftarrow \Theta_t$       [pool too small to trust]
3: end if
4: estimate the radio map distribution across  $\hat{x}_t = \Theta_t(x)$ ,  $\hat{x}_{\text{new}} = \Theta_{\text{new}}(x)$ ,  $x \in V$ ;
5:  $\mathcal{E}_t \leftarrow \text{RMSE}(\hat{x}_t, x)$ ,  $\mathcal{E}_{\text{new}} \leftarrow \text{RMSE}(\hat{x}_{\text{new}}, x)$ ,  $x \in V$ ;
6: if  $\mathcal{E}_{\text{new}} \leq \mathcal{E}_t - \varepsilon_{\text{gate}}$  then
7:    $\Theta_{t+1} \leftarrow \Theta_{\text{new}}$ ;      [confirmed under shift]
8:    $\bar{z}_i^2 \leftarrow 1 + \rho(\bar{z}_i^2 - 1) \quad \forall i$  [residuals to calibration]
9: else
10:   $\Theta_{t+1} \leftarrow \Theta_t$ ;
11: end if
12: return  $\Theta_{t+1}$ 

```

Credit Assignment

Efficient and stable policy updates with sparse reward

➤ Reward reconstruction

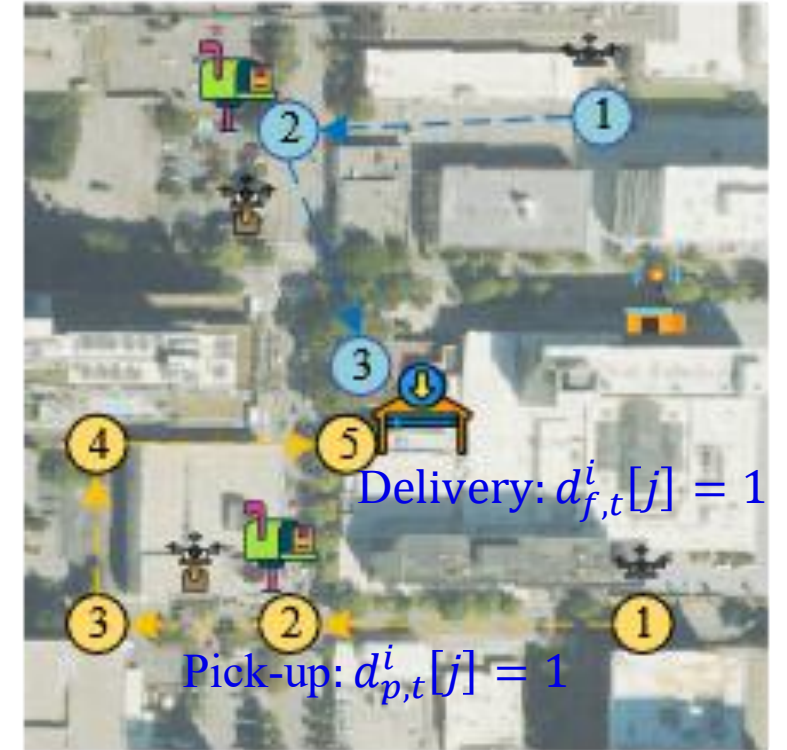
❖ Attribution of advantage

$$A_{i,t} = A_t - \lambda A_t^c = (A_t^r + A_t^i) - \lambda A_t^c, \quad r_{i,t} = r_t^g + r_t^i$$

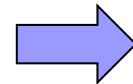
❖ Reward: only the terminal reward is sparse.

✓ Encourage pick-up and delivery.

✓ Stable policy gradient propagation.



$$\begin{aligned} r_t^g &= -1 + \overbrace{r_f d_{f,t}}^{\text{complete}} \\ r_t^i &= -\mu p_{x_{i,t}}^{\text{out}} + r_p d_{p,t}^i + r_d d_{d,t}^i \\ &\quad + \gamma \Phi_{d,i}(s_{t+1}) - \Phi_{d,i}(s_t) \end{aligned}$$



$$\begin{aligned} r_t^0 &= r_{i,t} - r_f d_{f,t} \\ r_{i,t}^0 &= \sum_{j \in m} \mathbb{I}(d_{p,t}^i[j] - d_{f,t}[j] = 1) \frac{r_{f,j}}{(t_{d,j} - t_{p,j} + W)t} \\ A_{i,t} &= \text{GAE}(r_t^0) + \sum_{l \in \{t_{d,j}, \dots, t_{p,j}\}} (\gamma \psi)^l r_{i,t+l}^0 - \lambda A_t^c \end{aligned}$$

Theoretical Analysis

□ Lemma

- Bounded reward and cost,
- Bounded, deterministic and data-independent prime dual updates,
- Exogenous-noise representation of approximation on constraints and RME model,
- The finite policy, action, observation and VAE params space are weak compact Borel space,
- The non-empty feasible set.

Theoretical Analysis

□ Theorem

- Existence: A stationary Markov optimum exists with fully measurable and monotone increasing mission completion.

- ❖ No continuity requirement of the transition, reward and cost.

$$\exists \pi^{\text{a}} \in \Pi, J_C(\pi^{\text{a}}) \leq \Gamma_c, J_R(\pi^{\text{a}}) = \sup_{\pi \in \Pi_{\text{feas}}} J_R(\pi)$$

- Convergence: bounded expectation.

$$\mathbb{E}\left[\frac{1}{K} \sum_{k=0}^{K-1} (J(\pi_k^{\text{a}}) - J(\pi_k))\right] \leq \rho_K = \frac{\lambda_{\max}}{\eta_{\lambda} K} - \gamma \sum_{j=0}^{K-1} \log \frac{r_{f,j}}{mW},$$

$$\frac{1}{K_{\text{eff}}} \sum_{k=k_0}^{K-1} [g(\lambda_k) - g(\lambda^{\text{a}})] \leq \frac{|\lambda_{k_0} - \lambda^{\text{a}}|^2}{2\eta_{\lambda} K_{\text{eff}}} + \frac{\eta_{\lambda} G^2}{2},$$

$$G \geq \sup_k |\hat{C}_k - \Gamma_c|, g(\lambda) = \max_{\pi} \{J_R(\pi) - \lambda(J_C(\pi) - \Gamma_c)\}$$

- Constraint Satisfaction: with proper $(\eta_{\lambda}, \lambda_{\max}, T)$, the dual ascent admits an explicit, non-asymptotic, one-sided certificate on the time-averaged constraint.

Settings

□ Simulation settings

- 2D wireless system: 4 drones deliver 16 mails in 256×256 pixel regions (8×8 regions).
- Dataset: RadioMapSeer (physics shift DPM, IRT2) and region shift (Atlanta downtown)
- Settings: 3 seeds 100 updates (20 updates for eval)

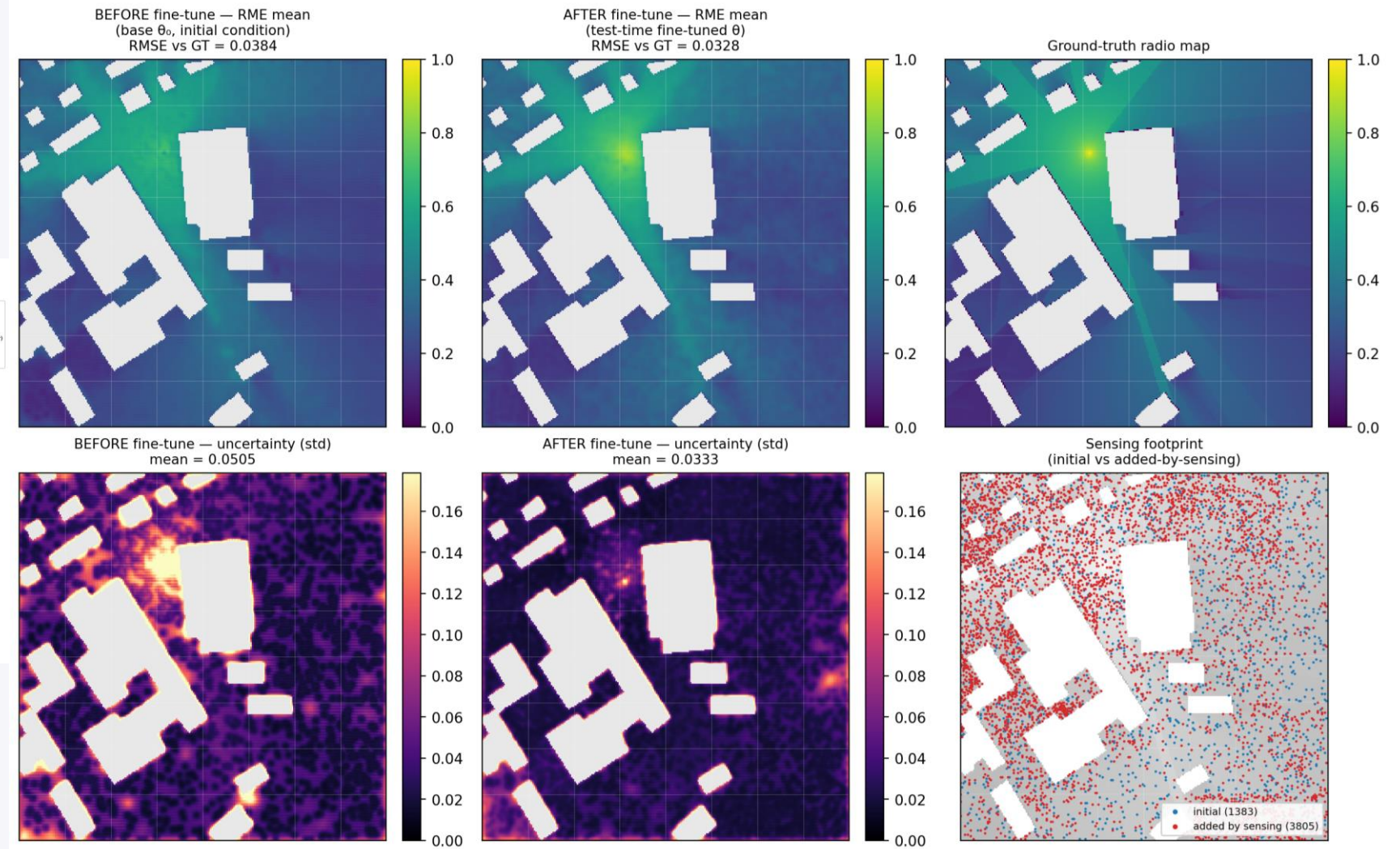
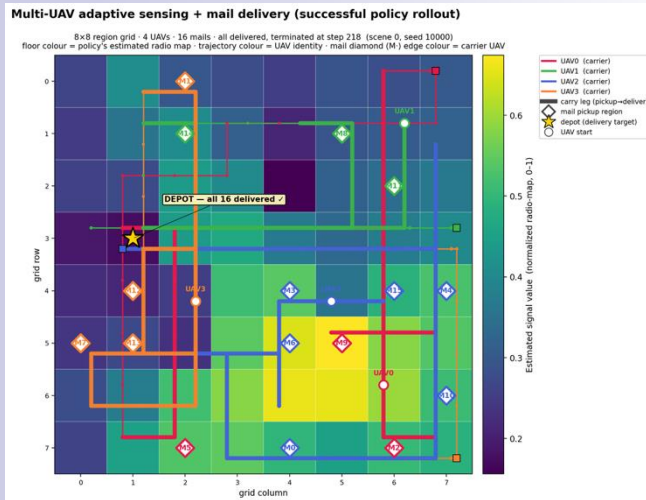
➤ Metrics:

- ❖ Success rate (robustness)
- ❖ Outage-MAE (safety)
- ❖ 90th percentile belief z-score $\bar{z}_{p90}$ (reliability)

Environment and task					
agents U	4	region grid	8×8	horizon T	300
mails m	16	actions	5	carriage capacity	unlimited
PPO and optimizer (Adam)					
clip radius ϵ	0.2	learning rate	3×10^{-4}	PPO epochs	4
minibatch size	1024	rollout length h	300	parallel envs	4
discount γ	0.99	GAE parameter ψ	0.95	entropy coef.	0.03
value-loss coef.	0.5	grad-norm clip	0.5	KL early stop	0.02
training updates	100	advantage norm.	per minibatch	reward scaling	running return std
Constraint and dual update					
budget Γ_c		1.0 $\rightarrow$ 0.15, first 10 updates			
initial dual λ_0	0	dual step η_λ	0.05	dual cap λ_{cap}	3.0
target floor τ_{abs}	0.03	shrink factor k	0.7	outage weight ω	0.02
truncated episodes		terminal cost included in the dual update			
Reward weights					
outage penalty μ	0.1	$\tau_p/\tau_d/\tau_f$	1/5/20	task weight w_{task}	3.0
distance α_{dist}	0.5	potential const. K	auto-sized	shaping discount γ_Φ	0.99
sensing reward weight		0 (constraint-priced only)			
Credit assignment (reference arm only)					
shaping window W	4	advantage cap	1.0		
Sensing and test-time adaptation					
STAY budget N_s	16	flight budget N_f	8	fit : held-out split	3:1
min held-out V_{min}	16	adaptation steps	20	adaptation lr	5×10^{-5}
anchor weight β	1.0	accept gate ϵ_{gate}	0.01	trigger κ_z	2.5
trigger hysteresis	0.2	trigger EMA α	0.1	queue depth	2
MC decodes M	30	outage threshold ϕ	0.30	min pixels / region	16
calib. cap Γ_m	100	calib. min count	8	calib. EMA	0.9
accept decay ρ	0.5				
Generative belief pretraining					
latent dimension	128	KL floor (free bits)	0.5 nat	β_{max}	0.25
KL annealing	linear, 20 cp.	batch size	32	learning rate	4×10^{-4}
LR warmup	5 cp.	epochs	100	reconstruction loss	MSE
aux. uncertainty weight	0.1	aux. decodes	4	aux. sub-batch	8
measurement ratio	0.5-5%				

Results

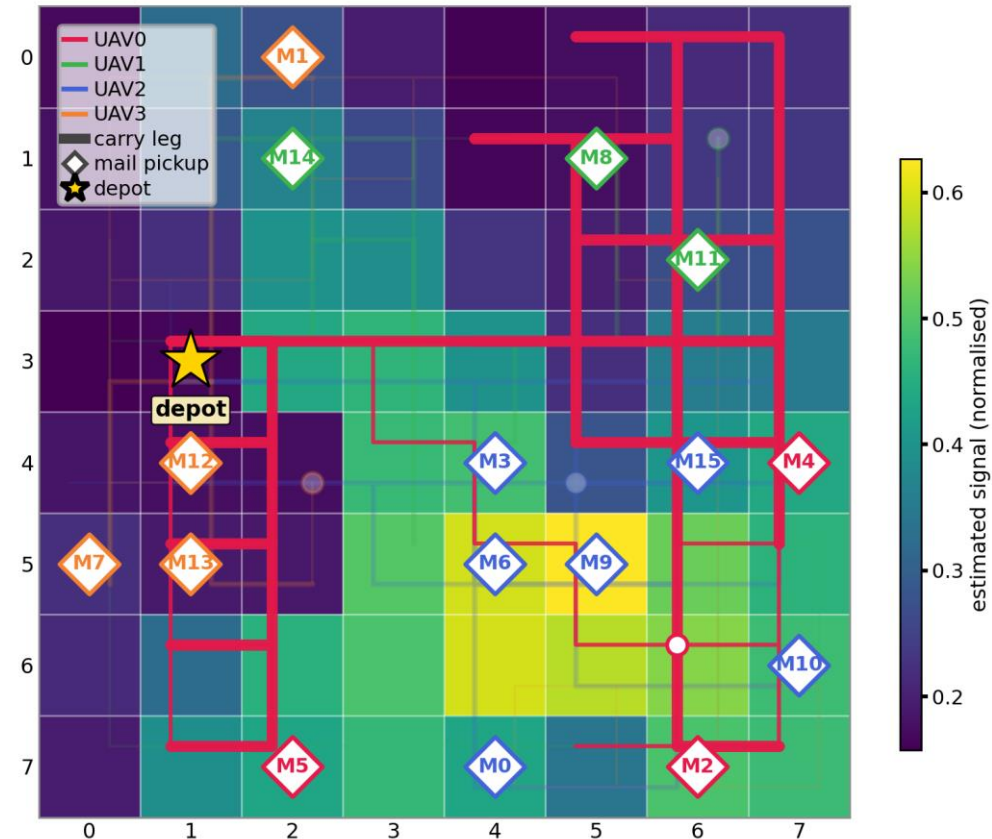
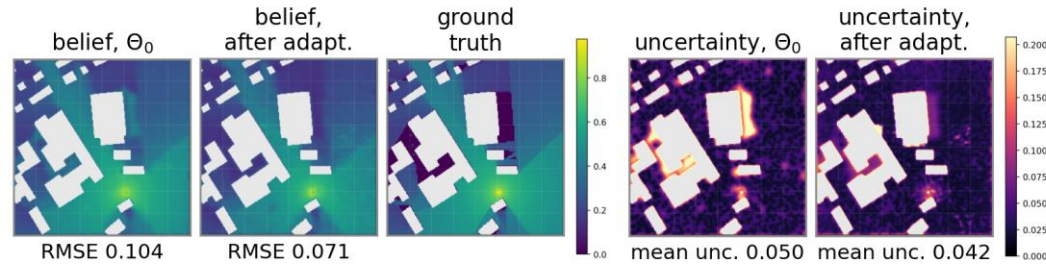
□ In-distribution



Results

□ Out of Distribution

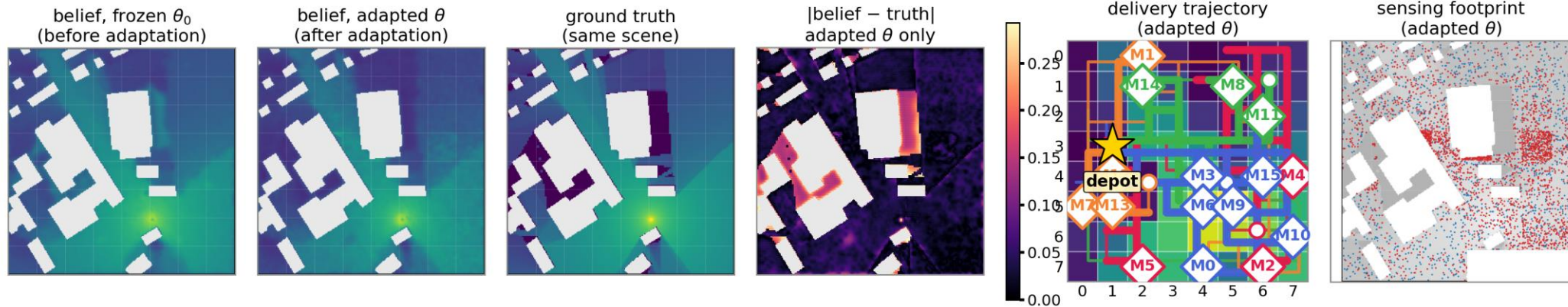
➤ Physics shift



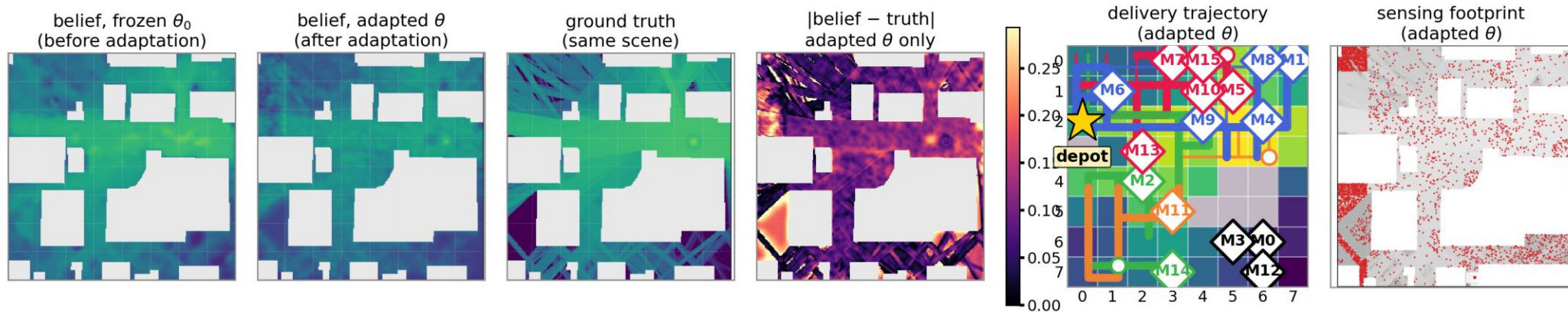
Results

□ Out of Distribution

shifted physics



unseen region



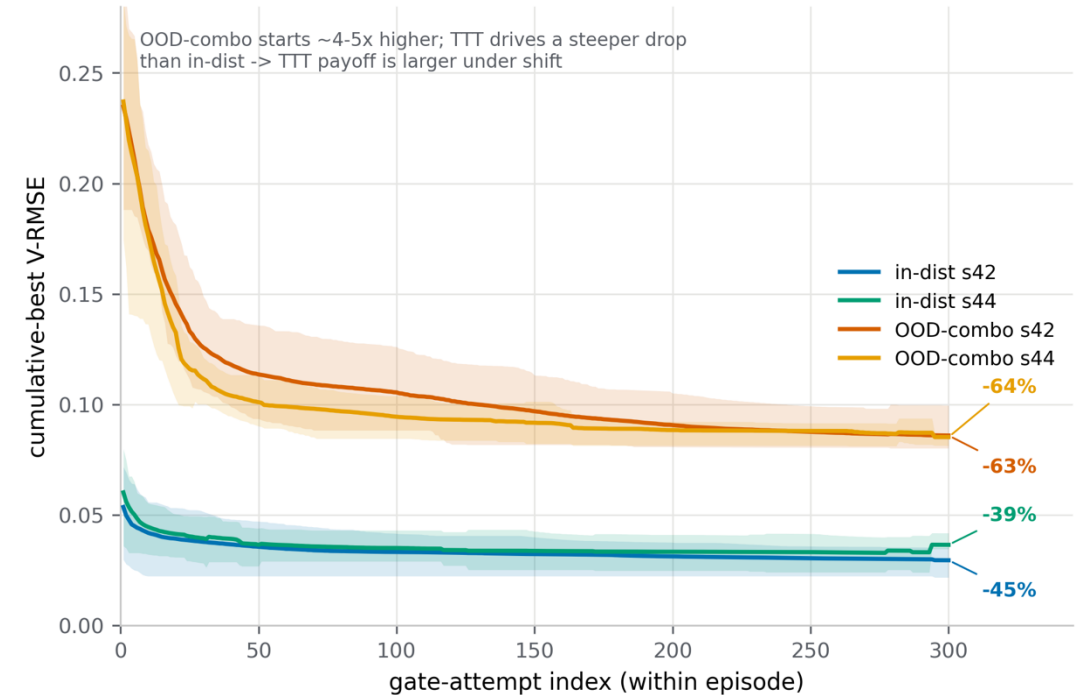
Results

□ Comparison

Table 2: Comparison on $m = 16$. Each baseline leverages STAY action and model fine-tuning under RME adaptation.

Baseline	SR				O-MAE				$\bar{z}_{p90}$			
	In-dist.	IRT2	carsIRT2	RMATL	In-dist.	IRT2	carsIRT2	RMATL	In-dist.	IRT2	carsIRT2	RMATL
MAPPO	0.580	0.608	0.566	0.663	0.089	0.127	0.130	0.080	1.39	3.02	3.50	3.12
	+0.033	+0.033	+0.018	+0.049	+0.016	+0.001	+0.010	+0.009	+0.45	+0.86	+0.44	+0.13
GiGPO	0.517	0.506	0.503	0.667	0.088	0.116	0.119	0.079	1.21	3.25	3.50	2.85
	+0.152	+0.168	+0.247	+0.285	+0.008	+0.014	+0.027	+0.001	+0.15	+0.67	+1.59	+0.56
AREL	0.844	0.732	0.784	0.897	0.091	0.135	0.129	0.074	1.46	3.76	3.43	2.79
	+0.186	+0.201	+0.199	+0.146	+0.015	+0.035	+0.035	+0.001	+0.40	+1.75	+1.75	+0.69
Ours	0.798	0.751	0.810	0.945	0.084	0.128	0.115	0.07	1.20	3.28	2.57	2.72
	+0.07	+0.182	+0.092	+0.013	+0.009	+0.023	+0.012	+0.001	+0.17	+1.36	+0.85	+0.82

Within-episode V-RMSE self-improvement (cumulative best)



Future Work

□ Agentic AI applications

- Improve C-MARL to agentic RL: introduce LLMs, remove task-oriented reward design
- Theoretical improvement of credit assignment
 - ❖ Hypothesis testing & counterfactual inference
 - ❖ Policy entropy optimization
- Cooperation and competition design (partial observation)
- Multimodal wireless knowledge

Thanks!